a) Use Eqs. 2.8 to show that

$$\alpha^*\alpha = \beta^*\beta = \gamma^*\gamma = \delta^*\delta = \frac{1}{2}.$$

b) Use the above result and Eqs. 2.9 to show that

$$\alpha^*\beta + \alpha\beta^* = \gamma^*\delta + \gamma\delta^* = 0.$$

c) Show that $\alpha^*\beta$ and $\gamma^*\delta$ must each be pure imaginary.

If $\alpha^*\beta$ is pure imaginary, then α and β cannot both be real. The same reasoning applies to $\gamma^*\delta$.

2.5 COUNTING PARAMETERS

It's always important to know how many independent parameters it takes to characterize a system. For example, the generalized coordinates we used in *Volume I* (referred to as q_i) each represented an independent degree of freedom. That approach freed us from the difficult job of writing explicit equations to describe physical constraints. Along similar lines, our next task is to count the number of physically distinct states there are for a spin. I will do it in two ways, to show that you get the same answer either way.

The first way is simple. Point the apparatus along any unit 3-vector[2] $\hat{n}$ and prepare a spin with $\sigma = +1$ along that axis. If $\sigma = -1$, you can think of the spin as being oriented along the $-\hat{n}$ axis. Thus, there must be a state for every orientation of the unit 3-vector $\hat{n}$. How many parameters does it take to specify such an orientation? The answer is of course two. It takes two angles to define a direction in three-dimensional space.[3]

Now, let's consider the same question from another perspective. The general spin state is defined by two complex numbers, α_u and α_d. That seems to add up to four real parameters, with each complex parameter